# **pyhydra**: an open, modular Python library for reproducible hydroclimatic modelling and probabilistic flood-risk analysis

Salvador Navas[a,*], Manuel del Jesus[b]

[a] *Hidralab Ingeniería y Desarrollos, S.L., ETSI Caminos UCLM, Laboratorio de Hidráulica UCLM, Av. Camilo José Cela s/n, Ciudad Real, 13005, Castilla-La Mancha, Spain*

[b] *Departamento de Ciencias y Técnicas del Agua y del Medio Ambiente, Universidad de Cantabria, Avenida de los Castros 44, Santander, 39005, Cantabria, Spain*

## Abstract

Flood-risk studies chain data acquisition, extreme-value and copula statistics, stochastic rainfall generation, climate bias correction and hydrological and hydraulic simulation, usually through ad-hoc scripts that are hard to reuse and reproduce. **pyhydra** is an open, modular Python library that consolidates this chain into fifteen sub-modules: harmonised access to public hydro-climatic data, statistical and copula tools, stochastic generators, bias correction and downscaling, automation of HEC-HMS, HEC-RAS, SFINCS and SWAT+, and uncertainty propagation, including a pilot-based rule that decides whether an expensive solver can be emulated within a fixed simulation budget. It is illustrated through four workflows at three Spanish sites; external-engine outputs are archived and third-party adoption is still to be shown. **pyhydra** is GPL-3.0-licensed, installable from GitHub (not yet on PyPI) and archived on Zenodo; the companion platform HYDRA exposes its notebooks in a browser.

*Keywords:* flood risk, reproducible research, hydroclimatic data harmonisation, stochastic rainfall generation, extreme value statistics, uncertainty quantification

*Corresponding author.

*Email address:* salvador.navas@hidralab.com (Salvador Navas)

*URL:* https://orcid.org/0000-0001-7439-1743 (Salvador Navas), https://orcid.org/0000-0003-0703-8960 (Manuel del Jesus)

## Required Metadata

## Current code version

| Nr. | Code metadata description | Metadata |
|---|---|---|
| C1 | Current code version | v0.3.1 |
| C2 | Permanent GitHub link to code/repository used for this code version | `https://github.com/navass11/pyhydra/tree/v0.3.1`; archived at `https://doi.org/10.5281/zenodo.23599934` |
| C3 | Legal Code License | GNU General Public License v3.0 only (GPL-3.0-only) |
| C4 | Code versioning system used | git |
| C5 | Software code languages, tools, and services used | Python ≥3.9; continuously tested on Python 3.9–3.12 (section 2.2) |
| C6 | Compilation requirements, operating environments & dependencies | Not yet published on PyPI; install with `pip` directly from the GitHub repository in C2, optionally pinned to the tag in C1 (section 2.1). Core dependencies are declared in `pyproject.toml`; optional functionality is distributed through the `statistics`, `geospatial`, `models`, `notebooks` and `all` extras (section 2.1). External engines (HEC-HMS, HEC-RAS, SFINCS, SWAT+) are not bundled and must be installed on the host |
| C7 | If available Link to developer documentation/manual | `https://github.com/navass11/pyhydra/tree/v0.3.1` (README, `docs/` package/installation guide, and `notebooks/pilot_cases/`, which include the four illustrative examples of section 3, inspectable and executable from this repository without HYDRA, subject to the external-engine and project-data requirements listed in Table 3). The companion platform HYDRA additionally exposes the same notebooks through a browser at `https://github.com/navass11/HYDRA/tree/v0.2.2` |
| C8 | Support email for questions | salvador.navas@hidralab.com |

Table 1: Code metadata (mandatory)

## 1. Motivation and significance

Flood-risk assessment usually requires combining several disjoint stages: acquiring precipitation, discharge, soil and climate-projection data from heterogeneous public services; fitting extreme-value and dependence models to

characterise the probabilistic structure of hazard; generating or selecting a tractable set of representative scenarios; correcting the bias of climate-model output; and finally running one or more hydrological and hydraulic engines to translate scenarios into physical impacts such as water depth, velocity or inundated area. In practice this chain is implemented as one-off scripts tied to a specific study, which makes results difficult to reproduce, audit or transfer to a different catchment, team or regulatory context – a recurring obstacle in flood-risk management, where under-resourced regions are disproportionately exposed to under-characterised hazard [1].

**pyhydra** packages a line of methodological work on impact-based flood-risk analysis into reusable software instead of study-specific code. The approach originates in a stochastic, continuous-simulation methodology for the Besaya River at Los Corrales de Buelna (Cantabria), which combined geostatistical event generation with probabilistic regression to derive return periods directly on hydraulic impacts rather than on design hyetographs [2]. Subsequent published work extended this line to spatial stochastic rainfall fields through the companion NEOPRENE library [3], a Gaussian-copula-plus-MaxDiss pipeline for urban flood-frequency analysis in Madrid [4], climate-change impact assessment on reservoir inflows [5] and Andean hydropower systems [6], and vine-copula-based multivariate design storms [7]. Each of these studies re-implemented overlapping pieces of the same statistical and modelling machinery. **pyhydra** consolidates that machinery into a single, tested, versioned codebase, so that a new case study is assembled from existing building blocks – data connectors, extreme-value fitting, stochastic generators, downscaling operators and model adapters – instead of being rewritten. **pyhydra** does not replace the hydrological/hydraulic engines it automates (HEC-HMS [8], HEC-RAS [9], SFINCS [10], SWAT+ [11]) or the public data services it connects to [12–15]; it is the reproducible integration layer that turns their combination into an auditable, end-to-end workflow. A companion platform, HYDRA, distributes that layer with a Docker environment and browser interface for transfer to non-specialist users, without being itself part of the code metadata in table 1.

Concretely, **pyhydra** addresses five recurring needs from the studies above: an installable, dependency-light scientific core; a modular, reproducible architecture usable axis-by-axis; harmonised access to hydro-climatic and hydrological data; automation of hydrological/hydraulic model execution; and propagation of uncertainty across the whole chain, from Monte Carlo ensembles to Bayesian posteriors – described in section 2 and demonstrated in sections 3 and 4. The companion platform HYDRA separately exposes this core through a web/API/Jupyter/Docker environment (section 2.1).

The contribution is therefore not a new probability distribution or physical

solver. It is the software architecture that makes these existing methods composable across the complete path from source data to impact-based frequency products, while retaining the native engines as independent, replaceable components. Statistical components can be used without a hydraulic model, and model adapters can be exchanged without rewriting the climate and uncertainty layers.

The closest open frameworks cover complementary but narrower parts of this path. CLIMADA targets global climate-risk assessment through hazard, exposure and impact functions [16]; RainyDay generates stochastic storms by transposing remotely sensed fields [17]; and HydroMT with wflow automates reproducible construction and execution of distributed hydrological models from global data [18, 19]. **pyhydra** does not seek to replace those tools. Its distinct scope is to combine multivariate extremes, stochastic generation, climate correction, professional hydrological/hydraulic adapters and impact-based return-period analysis in one inspectable workflow. Where appropriate, it consumes ecosystem tools such as HydroMT-SFINCS as dependencies rather than duplicating them.

## 2. Software description

### *2.1. Software architecture*

`pyhydra` is a pip-installable Python package with no web or infrastructure code, organised in four functional blocks – data sources, climate and statistics, model integration, and a UQ (uncertainty quantification) strategy layer – that communicate through standard data structures such as `pandas.DataFrame` for tabular series, `xarray.Dataset` for gridded data and GeoTIFF or `geopandas.GeoDataFrame` for spatial layers, so that a module in one block can be replaced or bypassed without breaking the others.

A separate, companion repository, `HYDRA`, consumes **pyhydra** as a dependency and adds an optional deployment layer on top of it: a JupyterLab environment, a FastAPI backend and an Astro/nginx web frontend, together with the Docker and Azure Container Apps configuration used to deploy them (fig. 1). This layer is what runs the browser-accessible illustrative examples in section 3; **pyhydra** itself has no dependency on it and can be used directly from any Python environment. **pyhydra** builds on code released under the GNU General Public License v3 and is therefore distributed under GPL-3.0-only; HYDRA is an independent project under the MIT licence.

Each **pyhydra** sub-module exposes its public API through its `__init__.py`, so the import pattern is always `from pyhydra.<block>.<module> import <name>`. OpenTURNS is a core dependency, installed by default with the core package, and provides the Gaussian-copula fitting used by

`climate.spatial_analysis` (table 2) and illustrated in the M-30 example of section 3; it is imported lazily inside the specific functions that need it, following the same lazy-import convention as the optional heavy dependencies below. Optional heavy dependencies (PyMC, GDAL, GeoPandas, pyvinecopulib) are imported lazily inside the functions that need them, so the package loads – and the parts of the API that do not need them run – even when those dependencies are absent; a user who only needs extreme-value analysis or bias correction can install **pyhydra** with `pip` alone. Everything else is grouped into four installable extras – `statistics` (PyMC, pyvinecopulib, lmoments3), `geospatial` (GeoPandas, rasterio, pykrige, earthaccess, hydromt_sfincs), `models` (hecdss, pySWATPlus, spotpy, NEOPRENE) and `notebooks` (seaborn, contextily, xlrd, openpyxl), plus a legacy `geo` extra for bare GDAL bindings and an `all` extra covering all five – so installing, for example, `pyhydra[statistics]` from the GitHub repository installs exactly what a given workflow needs. One extended dependency, CoSMoS_py (spatial random fields), is not yet published on PyPI and must be installed separately from its own repository. HYDRA's Docker Compose stack adds the extended scientific dependencies and orchestrates a JupyterLab container, a FastAPI backend and an nginx-served Astro frontend, built and deployed to Azure Container Apps by a GitHub Actions workflow, with project data mounted from a persistent file share rather than baked into the images.

### *2.2. Software functionalities*

Table 2 lists the fifteen sub-modules and their main capabilities. The statistical modules implement established methodology rather than novel formulations: bias correction follows standard delta/quantile-mapping approaches [20], regional frequency analysis follows the index-flood/L-moments framework [21], and dependence modelling follows standard copula theory [22] – consistent with the framing in section 1 that the contribution is architectural, not a new statistical method.

Four design principles guide the implementation: *modularity* (each block can be used independently – extreme-value analysis does not require geospatial or modelling dependencies); *reproducibility* (standardised notebooks, explicit parameters and data conventions preserve the evidence needed to regenerate a result, while identifying any required credentials, input projects or host-installed engines); *interoperability* (NetCDF/`xarray`, GeoTIFF, `pandas` and GeoJSON/`geopandas` as the common data currency between blocks and with external tools such as QGIS); and *operational transfer* (documentation, example notebooks and a reproducible Docker environment are treated as first-class deliverables, not optional accessories). A `pytest` suite of 299 tests in 19 test files exercises the statistical, modelling and UQ utilities most exposed to

regression. A GitHub Actions workflow (Ubuntu, core plus lightweight extended dependencies including PyMC) runs this suite on every push across a Python 3.9–3.12 matrix; the one test that depends on PyMC/pytensor is still marked `optional_dependency` and guarded to skip rather than fail if that stack is unavailable or, as happens with the older PyMC resolved on Python 3.9, incompatible with the installed NumPy. For the code version in Table 1 (commit `e96fbca`, 23 October 2026), this workflow reports 299/299 passed on Python 3.10–3.12 and 298 passed plus 1 skipped on 3.9, with 32% line coverage (31% on 3.9) – concentrated in the statistical and modelling-adapter code exercised directly by unit tests, lower in the data-connector and engine-automation modules whose behaviour is instead exercised, not formally verified, by the pilot-case notebooks of section 3 against real services and engines. Runs archived at `https://github.com/navass11/pyhydra/actions/runs/39978050001`.

*Reproducibility boundary.* Reproducibility is reported at the level supported by each workflow. Extreme-value fitting, copula analysis and bias-correction examples are executable from included or synthetic data in the container. Public-data downloads are repeatable after the user supplies the corresponding service credentials and records the query metadata. HEC-HMS, HEC-RAS, SFINCS and SWAT+ workflows additionally require the relevant engine and a model project; large pilot cases may also contain third-party data that cannot be redistributed. Figure 2 summarises the contract. HYDRA preserves notebooks, parameters, adapters and standard outputs for these conditioned workflows, but does not claim that containerising the Python layer redistributes proprietary executables or restricted datasets.

*Uncertainty propagation and analysis.* Rather than being confined to a single module, uncertainty quantification cuts across the four functional blocks: asymptotic, bootstrap and Bayesian posterior estimates on extreme-value parameters and return levels (`climate.time_series`); posterior propagation across the stations of a region and prediction variance from spatial interpolators (`climate.spatial_analysis`); and Monte Carlo roughness ensembles aggregated into spatial impact statistics with outlier screening (`modeling.hydraulic`) – the machinery exercised by the SFINCS–HEC-RAS demonstration case in section 3. The `uq` module is a dedicated decision layer sitting above these: given an ensemble and an expensive solver, it draws a nested maximum-dissimilarity pilot charged against a fixed evaluation budget `n_max`, measures that pilot's cross-validated emulability against a declared family of cheap emulators, and from that measurement chooses between completing a reduce-and-emulate design or falling back to direct Monte Carlo –

Table 2: **pyhydra** modules by functional block.

| Module | Block | Main capability |
|---|---|---|
| `data_sources.rainfall` | Data | ERA5, GPM, PERSIANN-CCS, AEMET, OGIMET, Meteostat |
| `data_sources.river_discharge` | Data | GloFAS, GRDC, USGS |
| `data_sources.climate_change` | Data | CMIP6 via CDS and ESGF |
| `data_sources.soils` | Data | SoilGrids download & USDA texture classification |
| `climate.time_series` | Climate | Event extraction, GEV/GPD fitting, NSE/KGE/PBIAS |
| `climate.spatial_analysis` | Climate | Regional frequency analysis, Gaussian copulas (OpenTURNS) / vine copulas (pyvinecopulib), IDW/kriging/GP interpolation |
| `climate.stochastic_generation` | Climate | NEOPRENE (point NSRP, spatial STNSRP), CoSMoS random fields |
| `climate.bias_correction` | Climate | Delta method, empirical/quantile-delta/scaled-distribution mapping |
| `climate.hybrid_downscaling` | Climate | Hydrograph classification, MaxDiss selection, k-NN/RBF flood-map reconstruction |
| `modeling.hydrology.hec_hms` | Modelling | Project generation, DSS I/O, SCE-UA calibration via spotpy |
| `modeling.hydrology.swat` | Modelling | SWAT+ input generation, execution, output parsing |
| `modeling.hydraulic.sfincs` | Modelling | Model setup, boundary conditions, batch execution |
| `modeling.hydraulic.hec_ras` | Modelling | 1D/2D project automation, DSS result extraction |
| `modeling.hydraulic.sensitivity` | Modelling | Manning Monte Carlo ensembles, spatial impact regression |
| `uq` | UQ | Pilot-based emulability diagnostic, MaxDiss design, reduce-and-emulate vs. Monte Carlo |

*Abbreviations:* GEV/GPD, Generalised Extreme Value / Generalised Pareto Distribution; NSE/KGE/PBIAS, Nash–Sutcliffe Efficiency / Kling–Gupta Efficiency / Percent Bias; IDW, Inverse Distance Weighting; GP, Gaussian Process; DSS, Data Storage System (HEC file format); SCE-UA, Shuffled Complex Evolution – University of Arizona; RBF, Radial Basis Function.

rather than fixing the choice by convention. A `StrategyDecision.margin` flags choices taken on a hair's breadth, and a companion sequential-design routine classifies each additional design point as improving, converged or deteriorating so that a flattened error curve is not mistaken for one that has turned upward.

### *2.3. Sample code snippets*

The snippet below chains three independent **pyhydra** components – bias correction, stochastic rainfall generation and bivariate copula analysis – each usable on its own:

```
from pyhydra.climate.bias_correction import BiasCorrection
from pyhydra.climate.stochastic_generation.point import NSRPModel
from pyhydra.climate.spatial_analysis.copulas import BivariateCopula

# 1. Bias-correct a regional climate model precipitation series
corrected = BiasCorrection.fit_transform(
    obs, hist, future, var='pr', method='qdm')

# 2. Calibrate and simulate a Neyman-Scott rectangular-pulse
# stochastic rainfall generator (NEOPRENE; 'models' extra)
model = NSRPModel(seasonality='monthly')
model.fit(observed_series)
synthetic = model.simulate(year_ini=2000, year_fin=2100)

# 3. OR-scenario 100-year iso-return-period contour of a compound
# rainfall-discharge event
copula = BivariateCopula(family='gumbel').fit(rain_max, discharge_max)
x_or, y_or = copula.joint_return_period(t=100, kind='or')
```

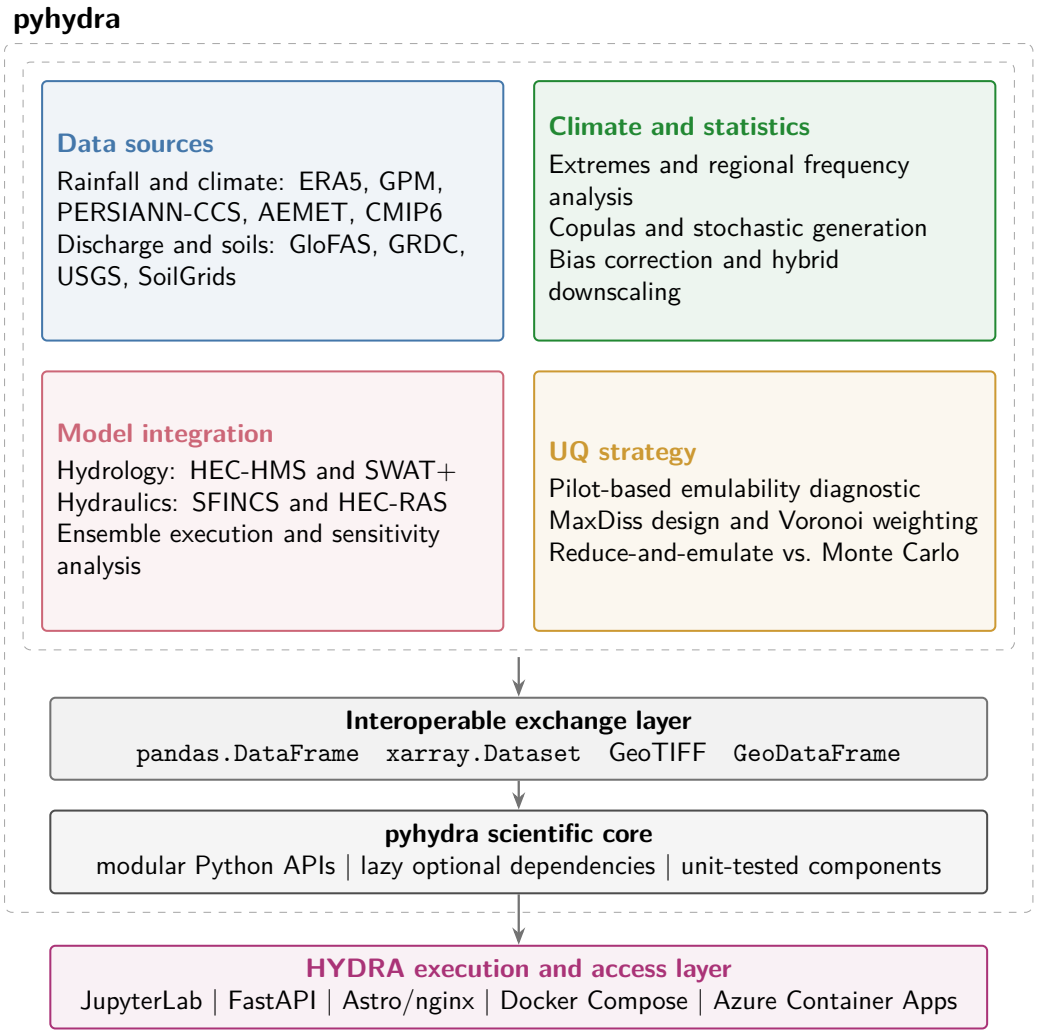



Figure 1: Software architecture. **pyhydra** provides four functional blocks of reusable Python components connected through standard exchange formats; HYDRA wraps the scientific core with reproducible execution, service and access layers. Colours identify functional roles and do not encode values.

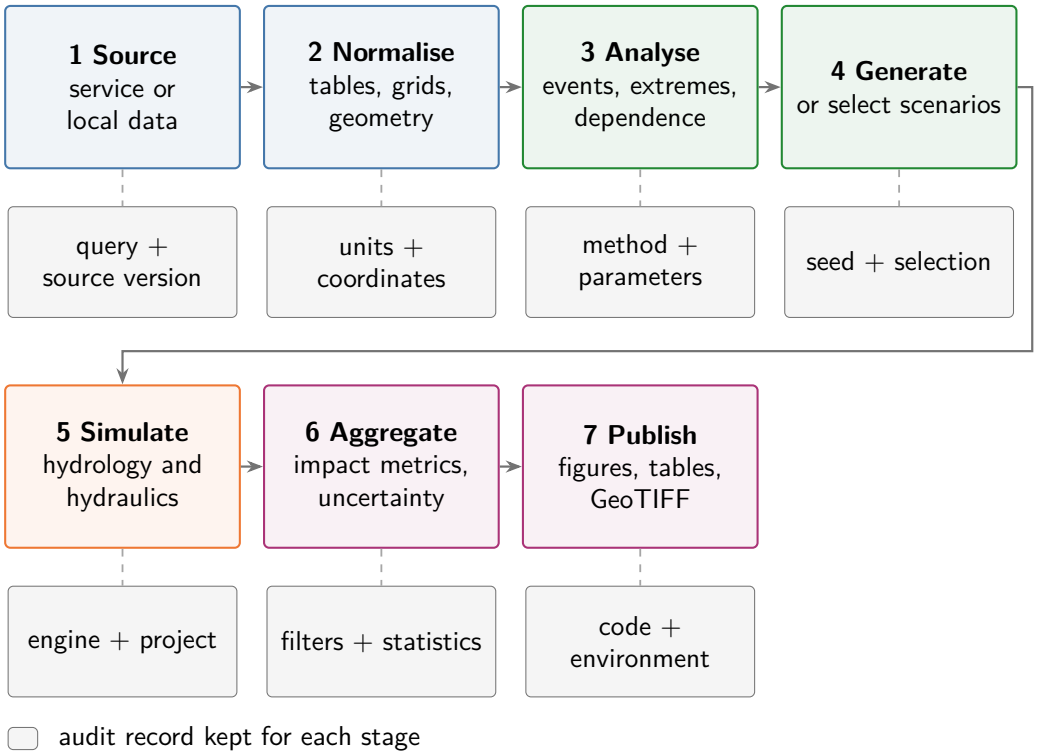



Figure 2: Reproducible workflow contract. Each transformation produces a scientific artefact and an audit record sufficient to identify its source, parameters, stochastic seed, external-engine configuration and software environment. Colours identify functional stages and do not encode values.

## 3. Illustrative examples

**pyhydra** has been exercised through four flood-risk workflows at three Spanish pilot sites, each reproduced as a notebook sequence under `notebooks/pilot_cases/` in the **pyhydra** repository itself (table 1, C7); the companion platform HYDRA exposes them in a browser. Table 3 traces each workflow to its permanent notebook sequence and states, per the reproducibility boundary defined above, what is required to re-run it. All 25 notebooks run to completion on the archived data: the nine Besaya notebooks skip their HEC-HMS/HEC-RAS steps when the engine is not installed, and the other 16 need no engine.

**Los Corrales de Buelna (Besaya River, Cantabria).** The founding case of the methodology [2, 25]: 42 years of continuous hydrological simulation (1970–2012) driven by 13 rain gauges, combined with geostatistical reconstruction of the rainfall series, copula-based generation of synthetic events and $k$-nearest-neighbour reconstruction of flood maps ($k = 6$, chosen by cross-validation) to estimate return periods directly on flood impacts rather than on design-storm discharge, in a valley-bottom town where the simulated floods may affect more than 13% of the population and 20% of the town's surface [25].

**M-30 Manzanares (Madrid).** Re-runs the statistical core of the multivariate flood-frequency study of Navas et al. [4] (fig. 3). In that study, 1,761 historical events from 17 rain gauges were classified by PCA + k-means. A Gaussian copula then generated approximately one million synthetic events, from which MaxDiss selected 625 representative events for HEC-HMS and steady-state HEC-RAS 1D (303 cross-sections). A k-NN regressor reconstructed depths across the synthetic ensemble under a compound Poisson frequency model. The resulting $T$=500-year design depth at Puente de Toledo exceeded the classical univariate-GEV estimate by 0.83 m (7.61 m vs. 6.78 m). The historical gauge events and the engine runs cannot be redistributed. The notebook sequence therefore fits **pyhydra**'s `FloodEventCopula` to an archived 10,000-event sample of the original synthetic catalogue. The fit uses an OpenTURNS Normal copula over 84 variables, with zero-inflated marginals for dry gauges. It regenerates 100,000 events that reproduce the sample's dry-gauge shares and its full rank-dependence structure (mean $|\Delta\rho| = 0.011$ over 3,486 variable pairs). It then selects 625 events with **pyhydra**'s MaxDiss in the full 84-dimensional space. The published design depths are read from the archived study outputs, not recomputed.

**Valencia's DANA (October 2024).** Implements an extreme-value fitting comparison for the record-breaking rainfall event that struck the Comunitat Valenciana on 29 October 2024, when AEMET's official post-event

Table 3: Traceability of the four illustrative workflows. Notebook paths are relative to `https://github.com/navass11/pyhydra/tree/v0.3.1/notebooks/pilot_cases/`. "Yes" marks inputs that are the authors' own or openly licensed. "Partial" marks cases where some inputs are third-party records (AEMET, SIAR, AVAMET, CHC), whose reuse and redistribution follow their providers' terms of use, or are project-specific and not redistributed here.

| Case | Path | Open data | External engine | Reproduction |
|---|---|---|---|---|
| Besaya | `los_corrales_buelna/` (9 nb.*) | Partial (projects and results on Zenodo [23]; gauge-derived series under provider terms) | HEC-HMS, HEC-RAS | Conditioned on the engines and on the gauge records |
| M-30 Manzanares | `m30_manzanares/` (6 nb.) | Partial (synthetic-catalogue sample and archived results on Zenodo [23]) | None (engine results archived) | Copula generation and MaxDiss re-run with **pyhydra**; design depths read from archived engine results |
| Valencia DANA | `valencia_dana/` (3 nb.†) | Partial (AEMET, SIAR and AVAMET series under provider terms; Zenodo [23]) | None | Reproducible for the point and regional (RFA) analyses with those records |
| SFINCS–HEC-RAS | `manning_rugosidades/` (7 nb.) | Yes (Zenodo dataset [24]) | SFINCS, HEC-RAS | Ensemble aggregation re-runs from the archived 2×1,000 outputs; re-simulation needs the engines |

*Steps 01–08 form the impact-frequency workflow; step 09 applies the `uq` decision layer to its MaxDiss design. †Steps 01–02 produce the results reported here; step 03 extends the regional analysis with a hierarchical Bayesian GEV (PyMC), using the legacy station matrix and the Stan reference results archived with the data.

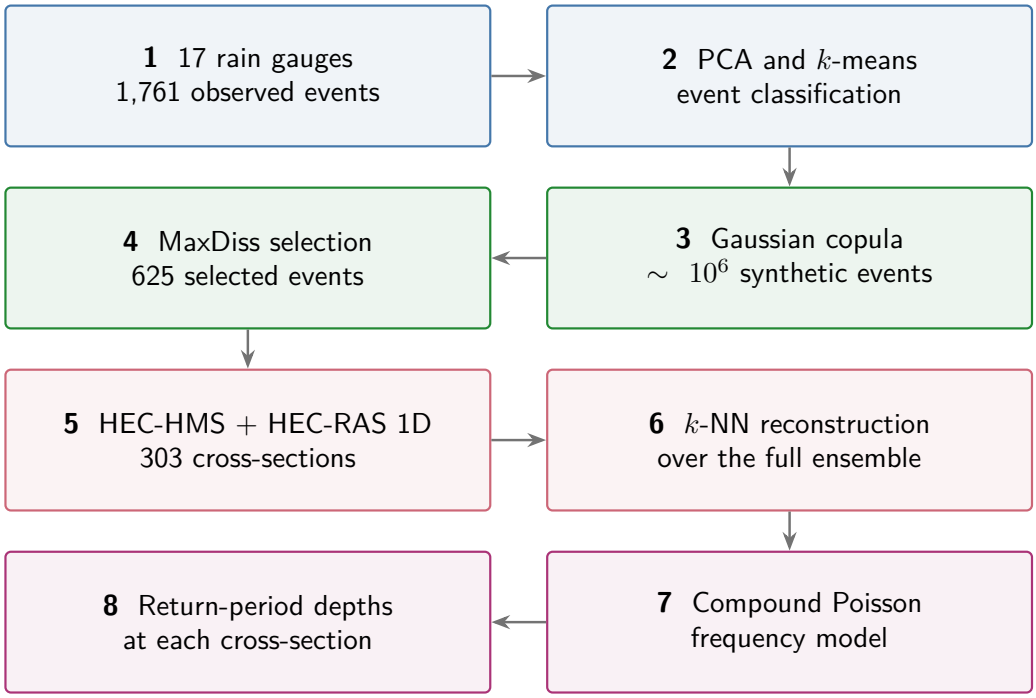



Figure 3: End-to-end pipeline of the M-30 Manzanares study [4]: from 17 rain-gauge series (step 1) to return-period depths at two control cross-sections, Puente de Toledo and Represa nº9 (step 8). The pilot notebooks re-run copula generation and MaxDiss selection with **pyhydra**; simulation, emulation and frequency outputs are taken from the archived study results. Blue denotes observations and event classification, green statistical generation and selection, orange physical simulation and emulation, and purple frequency outputs.

report [26] recorded new national single-station intensity records at Turís (station 8337X); the dataset used here, aggregated on calendar days, gives 710.8 mm/24h at that station. Two notebooks, downloadable in full with `pyhydra-get-data valencia_dana` and requiring no AEMET credentials at run time, fit GEV distributions at Turís with and without the event: point fits by three methods (MLE, L-moments, Bayesian/MCMC), and a regional frequency analysis (RFA; index-flood standardisation over all nine stations) by MLE and L-moments. Including the event raises the estimated $T$=100-year return level by a factor of 3.5–4.6 depending on method and scale, from 218–271 mm to 756–1,241 mm (fig. 4 and table 4). All values are recomputed with **pyhydra** v0.3.1. They are point estimates from a short annual-maximum record in which one record-breaking value dominates the upper tail, so the ratios show the sensitivity of the fit to that event, not calibrated return levels; sensitivity to the choice of stations is not assessed.

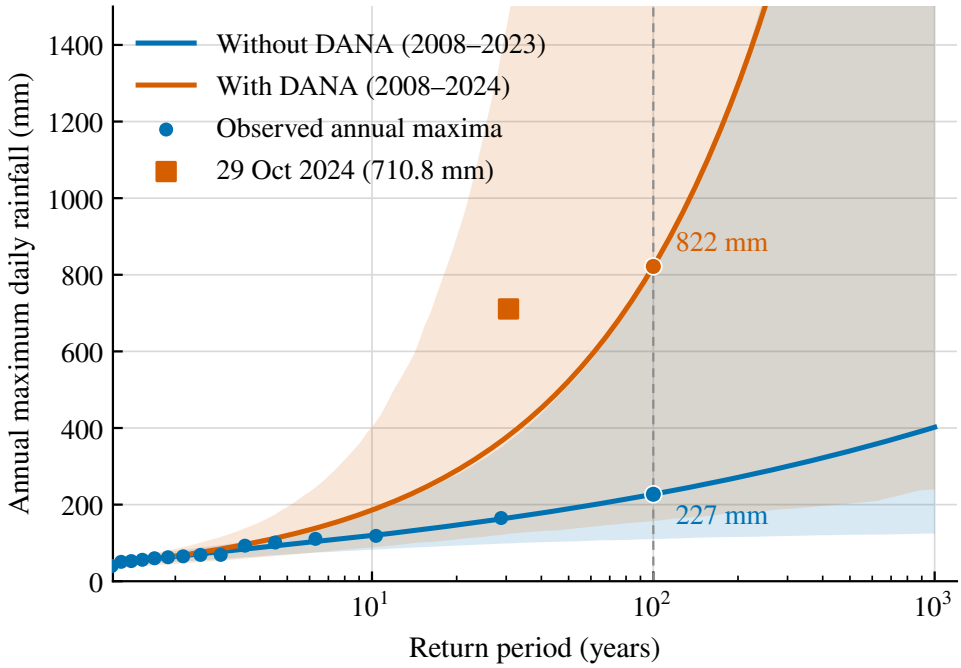



Figure 4: Turís (8337X): GEV return-level curves fitted by maximum likelihood without (16 years) and with (17 years) the 29 October 2024 event; bands: 95% Fisher intervals; dots: empirical positions (square: the 2024 record, 710.8 mm). The shape parameter rises from $\xi$=0.22 to 0.67 and the $T$=100-year level from 227 to 822 mm (table 4). The plotted values are those printed by the notebooks.

Table 4: $T$=100-year return level (mm) at Turís (8337X), without and with the 29 October 2024 event.

| Scale | Method | Without | With | Ratio |
|---|---|---|---|---|
| Point | Bayesian/MCMC | 258.5 | 932.6 | 3.6 |
| Point | L-moments | 218.0 | 755.6 | 3.5 |
| Point | MLE | 227.0 | 821.8 | 3.6 |
| Regional (RFA) | L-moments | 252.1 | 1,012.3 | 4.0 |
| Regional (RFA) | MLE | 271.3 | 1,241.3 | 4.6 |

**SFINCS–HEC-RAS uncertainty comparison (Besaya River).** A fourth demonstration case exercises the uncertainty-propagation machinery of section 2.2 across two independent 2D hydraulic engines: `generate_manning_combinations` draws 1,000 Monte Carlo roughness realisations over nine land-use classes from literature distributions, the same ensemble is run through both SFINCS and HEC-RAS, and `load_sfincs_ensemble`/`load_flood_ensemble` together with

`spatial_stats` and `manning_flood_regression` aggregate each engine's inundation output into comparable spatial impact statistics (figs. 5 and 6). This section presents only the **pyhydra** ensemble-generation and aggregation workflow, on one engine, as a software demonstration; the resulting two-engine scientific comparison between SFINCS and HEC-RAS is out of scope here and reported separately. Model configurations (including terrain and mesh), the 1,000 SFINCS and 1,000 HEC-RAS outputs and the derived tables are archived openly on Zenodo [24].

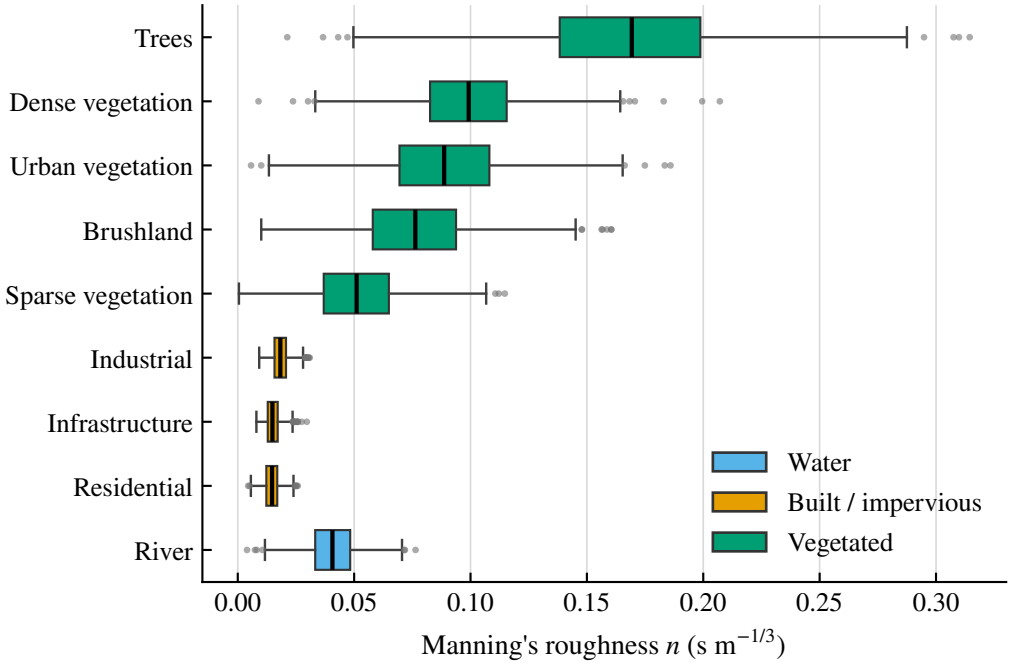



Figure 5: Input ensemble of the SFINCS–HEC-RAS demonstration case: the 1,000 Manning-roughness combinations (nine land-use classes sampled from literature distributions) that drove the simulations, grouped by land-cover type (water, built/impervious, vegetated) and ordered by median within each group. Boxes: interquartile range; line: median; whiskers: most extreme values within 1.5 times the interquartile range; dots: values beyond. 995 of the combinations gave valid runs, and the roughness of the river class is the one that controls the flooded area (fig. 6). Data from the Zenodo dataset [24].

The four workflows follow complementary execution paths – continuous impact-frequency analysis with hydraulic ensembles (Besaya), rainfall, runoff and hydraulic modelling (M-30), statistical inference without an external solver (Valencia), and Monte Carlo roughness ensembles on two hydraulic engines (SFINCS–HEC-RAS) – and are not the only published applications of the underlying methodology: the same building blocks also support published work on Andean hydropower systems [6] and reservoir inflows under

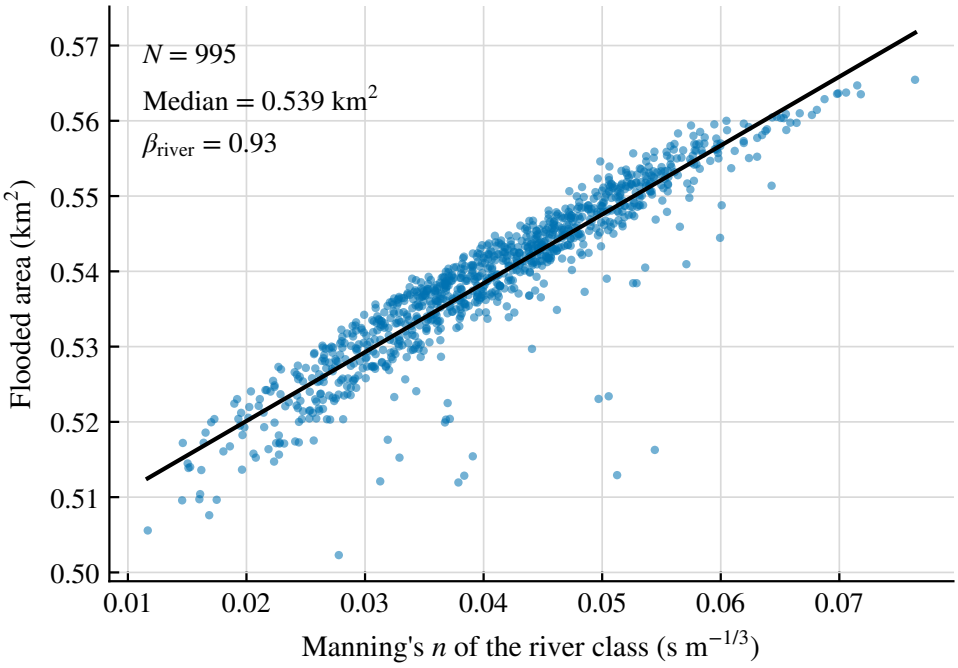



Figure 6: Output of the same case: flooded area of each of the 995 valid SFINCS runs (five numerically non-converged runs removed) against the Manning $n$ of the river class, from the archived per-run results [24]; line: least-squares fit. In a regression on the nine standardised classes the river carries $\beta$=0.93 and every other class $|\beta| \leq 0.08$ ($R^2$=0.87). This single-engine result documents the ensemble-aggregation workflow only; the two-engine comparison is out of scope here and reported separately.

climate change [5].

## 4. Impact

**pyhydra** formalises, as versioned and tested software, a line of flood-risk research that previously existed as independent, non-interoperable study codebases. Six published journal articles, produced by the authors' group, already rely on methods now consolidated in **pyhydra** [2–7]. Conference proceedings and manuscripts still under review are not counted here as impact evidence. This methodological track record predates **pyhydra** as a software product and shows that the underlying methods work; it is distinct from, and does not substitute for, evidence that the software itself has been adopted outside the authors' group.

Beyond consolidating methods that already existed as separate scripts, having them share a common ensemble and impact-statistics layer (section 2.2) also opens questions that were impractical to pose across four disconnected codebases: how differently two hydraulic engines propagate the same roughness ensemble into inundation extent, for which section 3 demonstrates the ensemble-generation and aggregation workflow and the comparison itself is reported separately; and how directly conditioning frequency analysis on spatial impacts, rather than on discharge or rainfall alone, shifts design estimates relative to the classical univariate route, as the published M-30 study quantifies (section 3).

As a software product, **pyhydra**'s impact so far is internal to the authors' group, reported as such rather than overstated. A new case study is now assembled from existing, unit-tested building blocks rather than rewritten – what let the four workflows of section 3 share one code base instead of four disconnected scripts. Applying the generic copula fitter to the real M-30 catalogue also exposed and fixed a defect that study-specific scripts would have hidden: with the default all-family search, heavy-tailed marginals were selected for zero-inflated gauge rainfall. Restricted families and zero-inflated marginals are now part of the tested API. Archiving **pyhydra** under the GPL-3.0 licence on Zenodo [27] gives each downstream publication a citable, versioned code reference, closing a gap that affected the earlier, unpublished study scripts. The companion platform HYDRA (section 2.1), archived separately [28] and reachable, as of 24 October 2026, at the URL given in its README[1], removes local-environment setup as a barrier to inspecting any of the four notebooks with the full scientific stack pre-installed.

Because the public GitHub release is recent, independent adoption metrics – downloads, external contributions, third-party citations, teaching or com-

[1]Ephemeral Azure Container Apps deployment; useful for evaluation but not a citable scholarly reference. The Zenodo records remain the durable citation for both components.

mercial use – are not yet available, and none are claimed here. **pyhydra** is used in ongoing doctoral and project research at IHCantabria – Universidad de Cantabria; broadening use beyond the authors' group is future work (section 5).

## 5. Conclusions

**pyhydra** turns a multi-year line of impact-based flood-risk research into modular, tested, GPL-3.0-licensed software: an installable scientific core with a modular, reproducible architecture; harmonised access to hydro-climatic and hydrological data; automation of hydrological and hydraulic models; and propagation and analysis of uncertainty across the whole chain. The companion platform HYDRA separately provides an optional web, API, Jupyter and Docker environment. Four workflows at three Spanish pilot sites – Besaya, M-30 Manzanares and Valencia – span fluvial, urban and extreme-rainfall flood risk. Together they show that the same modular components support markedly different study designs, from continuous stochastic simulation to multivariate copula-based frequency analysis and Monte Carlo ensembles run through two hydraulic engines. These workflows show what the components support, not third-party adoption; the M-30 design depths and HEC-HMS/HEC-RAS results are archived study outputs, not a validation of **pyhydra**. Ongoing work focuses on widening automated test coverage as new modules are added, adding adapters for further hydraulic/hydrological engines, and documenting a stable public API to lower the barrier for external contributions beyond the authors' own research group.
The present release also has explicit limits. Short records and rare extremes remain information-limited even under bootstrap, Bayesian or regional uncertainty quantification, and regional analysis still requires a defensible homogeneous region; stochastic generators do not by themselves resolve climate non-stationarity; emulators and MaxDiss selection reduce hydraulic-ensemble cost but cannot replace physical-model validation, and depend on the representativeness of the simulated subset; and reproducibility of externally executed models is conditioned by engine version, licences and project availability. HYDRA makes these dependencies explicit and auditable rather than removing them.

## CRediT authorship contribution statement

**Salvador Navas:** Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Manuel del Jesus:** Conceptualiza-

tion, Methodology, Resources, Supervision, Funding acquisition, Validation, Writing – review & editing.

## Funding

Manuel del Jesus acknowledges the support provided by Project PCI2024-153483 (WaMA-WaDiT, "Water Management and Adaptation based on Watershed Digital Twins"), funded by MICIU/AEI/10.13039/501100011033 and co-funded by the European Union. This research did not receive any other specific grant from funding agencies in the public, commercial or not-for-profit sectors. The funding bodies had no role in study design, data collection, analysis, interpretation, manuscript preparation or the decision to submit the article for publication.

## Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The **pyhydra** version described in this article is available through the versioned GitHub and Zenodo records in Table 1. The companion deployment platform HYDRA used to run the browser-accessible workflows in section 3 is an independent open-source project under the MIT licence, archived at [28]. The HEC-HMS and HEC-RAS projects and the processed data of the Besaya case, the synthetic-event sample and archived results of the M-30 case, and the series and results of the Valencia case are archived on Zenodo under CC-BY-4.0 [23]. The SFINCS–HEC-RAS model configurations, simulation outputs, tables and scripts are archived separately [24]. The 1,761 historical M-30 gauge events are third-party data and are not redistributed, and the hydrological and hydraulic engines (HEC-HMS, HEC-RAS, SFINCS, SWAT+) must be obtained from their developers. The Besaya and Valencia deposits include series derived from records of AEMET and the Confederación Hidrográfica del Cantábrico (Besaya) and of AEMET, SIAR and AVAMET (Valencia). These third-party records are not offered here as open data: their reuse and redistribution remain governed by the providers' terms of use. The pilot workflows also consume the public data services cited in the text.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation the authors used ChatGPT and Codex (OpenAI) and Claude Sonnet 5, Claude Opus 5.5 and Claude Sonnet 5.5 (Anthropic) for language editing, text revision, formatting, figure and plotting code, running and fixing the software and notebooks, and compiling the manuscript. The tools did not design the methodology or validate its scientific content. The authors reviewed and edited the output, checked the cited evidence and numerical values, and take full responsibility for the content of the article and the archived code.

## Acknowledgements

The authors thank the Confederación Hidrográfica del Cantábrico, the Agencia Estatal de Meteorología, the SIAR network and AVAMET for the hydrological and climate records used in the associated case studies, and the Instituto Geográfico Nacional for openly available cartographic products (orthophotography, digital terrain models, land cover and other mapping). Computationally intensive workflows used a Microsoft Azure academic subscription.

## References

[1] G. Di Baldassarre, A. Montanari, H. Lins, D. Koutsoyiannis, L. Brandimarte, G. Blöschl, Flood fatalities in Africa: from diagnosis to mitigation, Geophys. Res. Lett. 37 (22) (2010). doi:10.1029/2010GL045467.

[2] S. Navas, M. del Jesus, J. M. Sánchez, Evaluación y análisis del riesgo de inundación del río Besaya a su paso por Los Corrales de Buelna, Cantabria, Rev. Obras Públicas 3598 (2018) 61–72, in Spanish.

[3] J. Diez-Sierra, S. Navas, M. del Jesus, NEOPRENE v1.0.1: A Python library for generating spatial rainfall based on the Neyman–Scott process, Geosci. Model Dev. 16 (2023) 5035–5048. doi:10.5194/gmd-16-5035-2023.

[4] S. Navas, M. del Jesus, J. Martín, P. Sánchez, Aplicación de una nueva metodología para mejorar la estimación de las frecuencias de inundación en Calle 30 (Madrid), Ing. Agua 28 (4) (2024) 263–279, in Spanish. doi:10.4995/ia.2024.22293.

[5] S. Navas, M. del Jesus, SIMPCCe: una herramienta para el análisis de aportaciones a embalses ante escenarios de cambio climático, Ing. Agua 29 (2) (2025) 132–148, in Spanish. doi:10.4995/ia.2025.23217.

[6] M. del Jesus, J. Paz, S. Navas, E. Turienzo, J. Diez-Sierra, N. Peña, Efectos del cambio climático en el recurso hídrico de los países andinos, Ing. Agua 24 (4) (2020) 219–233, in Spanish. doi:10.4995/ia.2020.12135.

[7] D. Urrea Méndez, D. V. Gómez Rave, M. del Jesus, Multivariate design storms using vine copulas and Gaussian process regression, J. Hydrol. 677 (2026) 135762. doi:10.1016/j.jhydrol.2026.135762.

[8] W. Scharffenberg, Hydrologic modeling system HEC-HMS: User's manual, Tech. rep., US Army Corps of Engineers, Hydrologic Engineering Center (2016).

[9] G. W. Brunner, HEC-RAS river analysis system: 2d modeling user's manual, Tech. rep., US Army Corps of Engineers, Hydrologic Engineering Center (2016).

[10] T. Leijnse, M. van Ormondt, K. Nederhoff, A. van Dongeren, Modeling compound flooding in coastal systems using a computationally efficient reduced-physics solver: including fluvial, pluvial, tidal, wind- and wave-driven processes, Coast. Eng. 163 (2021) 103796. doi:10.1016/j.coastaleng.2020.103796.

[11] P. W. Gassman, M. R. Reyes, C. H. Green, J. G. Arnold, The soil and water assessment tool: Historical development, applications, and future research directions, Trans. ASABE 50 (4) (2007) 1211–1250. doi:10.13031/2013.23637.

[12] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, et al., The ERA5 global reanalysis, Q. J. R. Meteorol. Soc. 146 (730) (2020) 1999–2049. doi:10.1002/qj.3803.

[13] G. J. Huffman, D. T. Bolvin, D. Braithwaite, K.-l. Hsu, R. J. Joyce, C. Kidd, et al., Integrated multi-satellite retrievals for the Global Precipitation Measurement (GPM) mission (IMERG), in: Satellite Precipitation Measurement, Springer, 2020, pp. 343–353. doi:10.1007/978-3-030-24568-9_19.

[14] L. Alfieri, P. Burek, E. Dutra, B. Krzeminski, D. Muraro, J. Thielen, et al., GloFAS – global ensemble streamflow forecasting and flood early

warning, Hydrol. Earth Syst. Sci. 17 (2013) 1161–1175. doi:10.5194/hess-17-1161-2013.

[15] L. Poggio, L. M. de Sousa, N. H. Batjes, G. B. M. Heuvelink, B. Kempen, E. Ribeiro, et al., SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty, SOIL 7 (2021) 217–240. doi:10.5194/soil-7-217-2021.

[16] G. Aznar-Siguan, D. N. Bresch, CLIMADA v1: a global weather and climate risk assessment platform, Geosci. Model Dev. 12 (2019) 3085–3097. doi:10.5194/gmd-12-3085-2019.

[17] D. B. Wright, R. Mantilla, C. D. Peters-Lidard, A remote sensing-based tool for assessing rainfall-driven hazards, Environ. Model. Softw. 90 (2017) 34–54. doi:10.1016/j.envsoft.2016.12.006.

[18] D. Eilander, et al., HydroMT: Automated and reproducible model building and analysis, J. Open Source Softw. 8 (2023) 4897. doi:10.21105/joss.04897.

[19] W. J. van Verseveld, A. H. Weerts, M. Visser, et al., Wflow_sbm v0.7.3, a spatially distributed hydrological model: from global data to local applications, Geosci. Model Dev. 17 (2024) 3199–3234. doi:10.5194/gmd-17-3199-2024.

[20] C. Teutschbein, J. Seibert, Bias correction of regional climate model simulations for hydrological climate-change impact studies: Review and evaluation of different methods, J. Hydrol. 456–457 (2012) 12–29. doi:10.1016/j.jhydrol.2012.05.052.

[21] J. R. M. Hosking, J. R. Wallis, Regional Frequency Analysis: An Approach Based on L-Moments, Cambridge University Press, 1997. doi:10.1017/CBO9780511529443.

[22] C. Genest, A.-C. Favre, Everything you always wanted to know about copula modeling but were afraid to ask, J. Hydrol. Eng. 12 (4) (2007) 347–368. doi:10.1061/(ASCE)1084-0699(2007)12:4(347).

[23] S. Navas Fernández, M. del Jesus, pyhydra pilot-case data: Besaya (Los Corrales de Buelna), M-30 Manzanares and Valencia DANA [dataset], Zenodo, version 1.1.0 (2026). doi:10.5281/zenodo.23599126.
URL https://doi.org/10.5281/zenodo.23599126

[24] S. Navas Fernández, Á. Galán, M. del Jesus, Data and code for: From roughness uncertainty to model-configuration effects in 2D flood inundation models: a Monte Carlo comparison of SFINCS and HEC-RAS [dataset], Zenodo, version v9 (2026). doi:10.5281/zenodo.23104366.
URL https://doi.org/10.5281/zenodo.23104366

[25] S. Navas Fernández, Evaluación y análisis del riesgo de inundación del río Besaya a su paso por Los Corrales de Buelna, Cantabria, Trabajo de fin de máster, máster en ingeniería de caminos, canales y puertos, Universidad de Cantabria, in Spanish (2018).
URL https://hdl.handle.net/10902/10636

[26] Agencia Estatal de Meteorología (AEMET), Informe sobre el episodio meteorológico de precipitaciones torrenciales y persistentes ocasionadas por una dana el día 29 de octubre de 2024, Tech. rep., AEMET, Ministerio para la Transición Ecológica y el Reto Demográfico, prepared with information available as of 8 November 2024 (2024).
URL https://www.aemet.es/documentos/es/conocermas/recursos_en_linea/publicaciones_y_estudios/estudios/informe_episodio_dana_29_oct_2024_.pdf

[27] S. Navas Fernández, M. del Jesus, pyhydra: a modular Python library for hydrological and climate analysis, computer software, version 0.3.1 (2026). doi:10.5281/zenodo.23599934.
URL https://doi.org/10.5281/zenodo.23599934

[28] S. Navas Fernández, M. del Jesus, HYDRA: web platform, notebooks and deployment environment for pyhydra, computer software, version 0.2.2 (2026). doi:10.5281/zenodo.23617748.
URL https://doi.org/10.5281/zenodo.23617748

### Current executable software version

**pyhydra** v0.3.1 (commit `e96fbca`) is the executable software version described in this publication; its archived record and immutable GitHub tag link are provided in Table 1 and should be cited independently using the corresponding Zenodo record [27]. The companion deployment platform HYDRA used for the browser-accessible examples in section 3 is archived separately at v0.2.2 (commit `3f5515f`) [28].